Quantile-Based Detection of Ecological Thresholds and Sampling Sensitivity in Large Rivers

Abstract

Modeling pressure–response relationships between contamination gradients and zoobenthic communities is central to environmental assessment. Such relationships often exhibit constrained state spaces, producing structurally empty "forbidden" regions where certain pressure–response combinations are ecologically implausible. We propose a hierarchical computational framework that integrates Piecewise Quantile Regression (PQRM), Threshold Indicator Taxa Analysis, and wild bootstrap resampling to estimate these threshold-induced forbidden subspaces and quantify associated uncertainty. The approach extends inference beyond observed data by propagating parameter variability to model-level uncertainty. A subsampling–bootstrap scheme further evaluates minimum sampling effort required for stable threshold detection. Application to St. Clair–Detroit River data demonstrates how distribution-aware modeling improves conservative pollution classification under limited sampling conditions.

Keywords: Piecewise Quantile Regression, Change-Point Analysis, Resampling Methods, Nonparametric Inference, Ecological Modeling

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